

Lab Report

UI/UX Design for a Medicine Tracker

Prepared For

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Prepared By

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Dear Mr. Hossain Ali,

Subject: Software Development Proposal for a Medicine Reminder & Tracker App

Please find enclosed our detailed software development proposal for your kind consideration.

Our team specializes in designing modern, scalable, and user-centric mobile software solutions tailored to vital healthcare needs. We understand that accurate medication adherence, elder care, and health tracking are critical challenges for individuals and families within the Bangladesh healthcare system.

This proposal outlines an Android Mobile Application designed specifically to automate and optimize medication tracking. The system improves patient adherence and peace of mind for families through smart AI reminders, prescription scanning, stock alerts, and automated emergency notification systems.

With our focus on robust engineering, we emphasize long-term usability, local language accessibility, and system reliability. We appreciate your valuable time and look forward to your guidance on this project to contribute to digital healthcare solutions.

Yours sincerely,

Dumbledore Software Ltd.

EXECUTIVE SUMMARY

This lab report presents a comprehensive UI/UX design study for a **Medicine Reminder & Tracker App** — an Android-based mobile solution developed to address critical healthcare adherence challenges in Bangladesh. The system is designed to serve individual patients, elderly users, and their family caregivers.

The application enables users to manage medication schedules via manual entry or prescription scanning (QR/OCR), receive AI-driven smart reminders, track medicine stock levels, and trigger automated emergency alerts to family members if doses are missed consecutively. The app also provides a direct interface for emergency doctor consultation.

The design follows a user-centered approach with a strong emphasis on elderly-friendly ergonomics, Bangla language accessibility, and voice-assisted interactions. The prototype was developed using Figma, comprising 15 fully connected screens with prototype interactions linking every button to its corresponding destination screen.

The system includes a complete authentication flow, patient dashboard, active reminder interface with stock tracking, emergency alert simulation, and a dedicated voice/accessibility assistant panel.

Item	Detail
Project Name	Medicine Reminder & Tracker App
Target Platform	Android Mobile Application
Design Tool	Figma (UI/UX Prototype)
Total Screens	15 Screens
User Types	Patient / Caregiver, Family Member (Alert Recipient)

Item	Detail
Report Date	June 5, 2026

1) Project Overview

- **Project Title:** Medicine Reminder & Tracker App
- **Target Platform:** Android Phone Base
- **Core Approach:** Smart, Accessible Healthcare Utility

Project Purpose

The Medicine Reminder & Tracker App is designed to digitize, simplify, and automate daily medication adherence. The core objectives are:

1. Allow users to add medicine schedules manually or via prescription scanner (OCR/QR)
2. Provide AI-based adaptive reminder notifications to improve adherence
3. Automatically detect missed doses and trigger SMS alerts to family members
4. Track remaining medicine stock and send shortage warnings
5. Integrate voice reminders for elderly or visually impaired users
6. Offer a direct emergency doctor consultation interface

2) Feasibility Analysis

2.1 Technical Feasibility

The project leverages highly reliable, robust, and industry-proven development tools and modules:

- **Android Studio:** Provides a robust, natively optimized environment for building a smooth and responsive user interface tailored for Android mobile phones.
- **AI Algorithm:** Drives the smart reminder core, adapting intelligently to optimal notification patterns.
- **SMS API Integration:** Guarantees dependable, instant routing of critical alerts to family members when emergency thresholds are crossed.
- **QR / OCR Scanner Integration:** Powers the prescription scanner feature, facilitating automated extraction and entry of medicine details.

2.2 Operational Feasibility

- **User Accessibility:** Features an incredibly simple, clutter-free user interface purposefully designed to be elderly-friendly.
- **Localized Experience:** Supports the Bangla language natively to break down accessibility barriers for the local population.
- **Audio Assistance:** Incorporates voice reminders to assist users with visual impairments or those who struggle with reading text notifications.
- **Societal Alignment:** Perfectly adapted for the Bangladesh family system by offering a family alert feature that loops in caregivers seamlessly.

2.3 Economic Feasibility

- **Development & Infrastructure Costs:** Expenses are limited to standard Android application development, cloud hosting/database storage for user profiles, and active SMS gateway API packages.
- **Value Generation:** High utility return for elderly patients and busy households, lowering accidental health risks due to non-adherence.
- **Monetization & Revenue:** Sustainable operation achieved by integrating short, relevant healthcare and wellness advertisements within the application platform. Highly marketable for potential integration with hospitals and local pharmacies.

3) Cost-Benefit Analysis (CBA)

The following table outlines the estimated development and operational costs against the projected financial benefits and revenue generation over a two-year period:

Item	Cost (BDT)	Benefit (BDT)
UI/UX Design (Figma – 15 screens)	150,000	—
User Testing & Elderly Accessibility Iteration	60,000	—
Development Cost Saving (rework reduction)	—	450,000
Reduced Post-Launch Bug Fixes (design-related)	—	180,000
Total	210,000	630,000

Benefit – Cost Ratio (BCR) Calculation

To evaluate the economic feasibility of the project , the benefit cost ratio (BCR) is calculate by diving the total projected benefits by the total expected cost:

$$\text{BCR} = \frac{\text{Total Expected Benefits}}{\text{Total Expected Costs}}$$

$$\text{BCR} = \frac{630,000}{210,000} \approx 3$$

Final Decision

Since the Benefit-Cost Ratio is significantly greater than \$1.0\$, the financial return outweighs the initial and operational investments. Therefore, the project is economically highly feasible and should be accepted for implementation.

4) 3. Screen Design & Description

The Medicine Reminder App prototype consists of 15 screens organized into 3 rows on the Figma canvas — Onboarding/ Auth Flow, Patient (Core) Flow, and Emergency/Family Flow — allowing a complete user journey overview.

3.1 Authentication & Onboarding Flow (Screens 01–05)

Screen 01 – Welcome / Splash Screen

Entry point with role selection and brand identity.

- Full-screen calming gradient (soft blue to teal)
- App logo (pill + bell icon)
- Role cards: **Patient / Caregiver** and **Family Member (Alert Recipient)**
- `Get Started` → primary button
- `Already have an account? Sign In` link

Screen 02 – Patient Login

Login screen for patients/caregivers.

- Soft gradient header with app name
- Phone Number / Email and Password fields
- `Forgot Password?` Link
- `Sign In` button → navigates to Patient Dashboard
- `Continue with Google` option
- `Create New Account` link → navigates to Register screen

Screen 03 – Patient Registration

New user registration with health profile setup.

- Full name, phone number, date of birth
- Emergency contact name & phone number (family member)
- Primary language selector (Bangla / English)
- Register & Continue button → navigates to Dashboard

Screen 04 – Family Member Login

Dedicated login for family members who receive alerts.

- Distinguishable orange accent header (different from patient blue theme)
- Phone number / Email, Password, optional **Patient ID to link**
- Sign In as Family button → navigates to Family Dashboard (screen 14)

Screen 05 – Voice/Assistance Setup

Accessibility configuration screen (shown once after first login).

- Toggle for **Voice Reminders** (On/Off)
- Select voice speed (Slow / Normal)
- Toggle for **Bangla Voice Prompts**
- Save & Go to Dashboard button

3.2 Patient Core Flow (Screens 06–10)

Screen 06 – Patient Dashboard (Home)

Main screen showing today's medications, adherence summary, and stock alerts.

- Header with patient name, greeting, and notification bell

- **Adherence card:** Today's status (e.g., "3 of 4 taken" with progress ring)
- **Active Reminders list:** Medicine name, time (e.g., 8:00 AM), dosage, Taken / Skip buttons
- **Stock Alert banner:** "Paracetamol – only 2 doses left" with Refill link
- **Missed Dose warning** (if applicable) in red
- Bottom navigation: Home | Medicines | Stock | Emergency | Profile

Screen 07 – Add Medicine (Manual Entry)

Form to add a new medication.

- Medicine name, dosage (e.g., 500mg)
- Time picker (e.g., 8:00 AM, 2:00 PM, 8:00 PM)
- Start date, End date (optional)
- Stock quantity (e.g., 30 tablets)
- Create Reminder button → saves and returns to dashboard

Screen 08 – Prescription Scanner (OCR/QR)

Scan prescription to auto-fill medicine details.

- Camera viewfinder overlay
- Upload from Gallery alternative button
- Scan Now button → processes image
- Extracted data preview: Medicine Name, Dosage, Frequency, Duration
- Auto-fill & Save button → navigates to confirmation screen

Screen 09 – Medicine Stock Tracker

Full inventory of all medicines with stock levels.

- List of medicines with percentage bar (Green for >30%, Orange for <30%, Red for <10%)
- Reorder button next to low-stock items
- Add to Shopping List for pharmacy visit
- Low-stock warning banner at the top

Screen 10 – Patient Profile & Settings

User account and app settings.

- Avatar with patient initials, name, emergency contact
- Account info: phone, date of birth, linked family member
- Settings: Notification preferences, Voice Reminder toggle, Bangla/English switch
- Emergency Contact Settings (add/edit family member number)
- Log Out button → navigates back to Welcome screen

3.3 Emergency & Family Flow (Screens 11–15)

Screen 11 – Emergency Alert Simulation & Doctor Contact

Critical safety screen for emergency situations.

- Large red SOS – Emergency Alert button (sends SMS to family immediately)
- Call Emergency Doctor button (direct dial to configured physician)
- **Missed Dose Consecutive Alert** panel: shows last missed dose, warning status, and Notify Family Now button
- I'm Okay – Cancel Alert button (if accidental trigger)

Screen 12 – AI Reminder Settings

Configure the smart reminder AI behavior.

- Adaptive Reminder toggle (AI learns best times based on user response)
- Quiet hours setting (e.g., 10 PM – 7 AM no notifications)
- Escalation: if dose not taken after 30 minutes → send push notification; after 60 minutes → SMS to family
- Preview of current AI schedule

Screen 13 – Missed Dose History & Report

Log of all missed and taken doses.

- Calendar view with color-coded days (Green = all taken, Orange = 1 missed, Red = >1 missed)
- List of missed doses with date, medicine, time, and action taken (alert sent or not)
- Share Report with Doctor button (PDF export)

Screen 14 – Family Member Dashboard

View for family members to monitor their loved one's adherence.

- Header: "Monitoring: [Patient Name]"
- **Today's Status:** "All doses taken" or "1 dose missed at 2:00 PM"
- **Day Adherence Chart** (bar chart with %)
- **Recent Alerts:** list of SMS notifications sent (date, reason)
- Send Reminder to Patient button (sends push notification to patient's phone)
- Contact Emergency Doctor button

Screen 15 – Voice Reminder Assistant (Full Screen)

Dedicated screen for hands-free voice interaction.

- Large microphone icon with "Listening..." animation
- Voice commands: "Read my reminders", "Mark morning dose as taken", "Check stock of insulin", "Call family"
- Text-to-speech confirmation after each command
- [Back to Dashboard](#) button

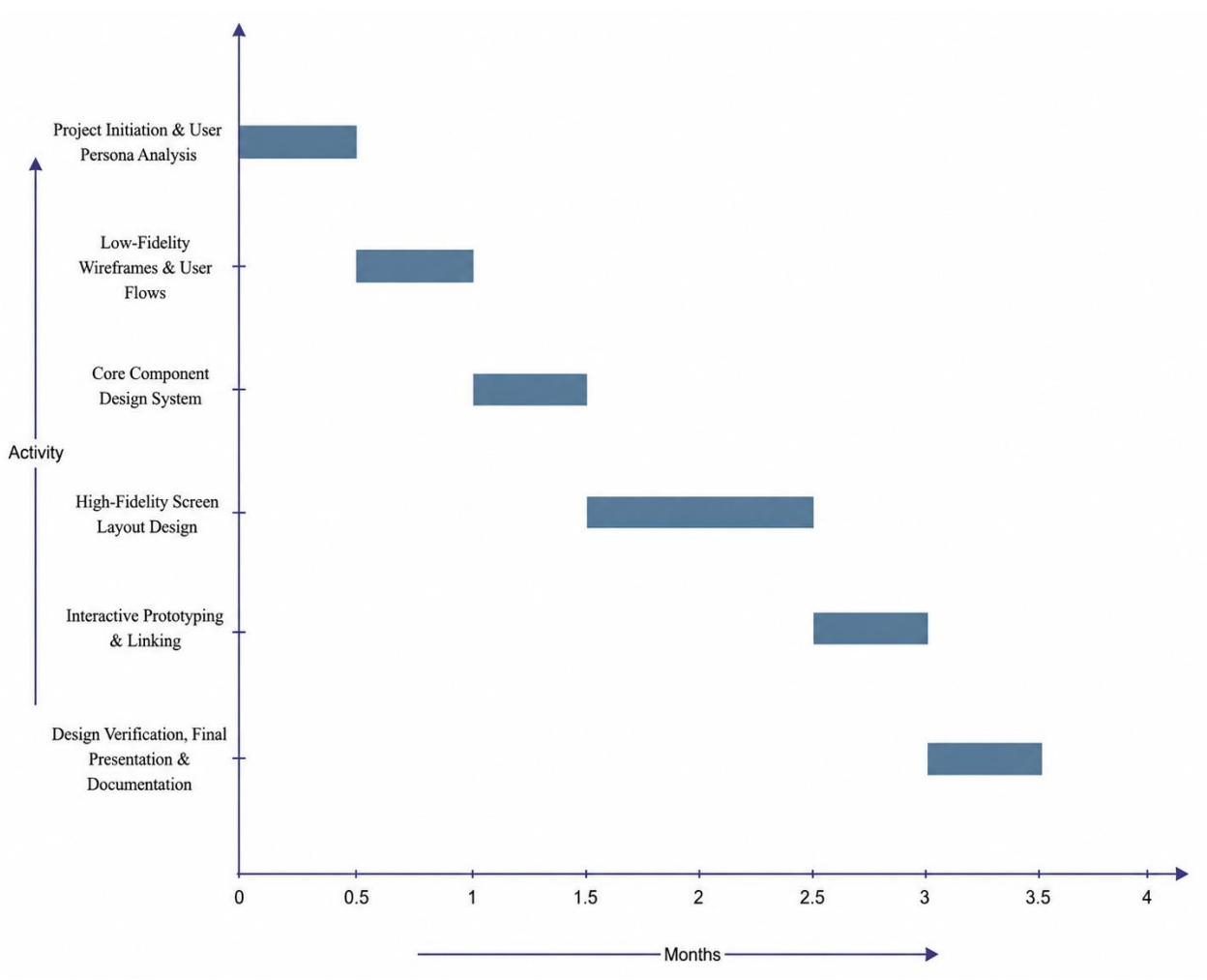
5) Sequence of Activities & Schedule

The following table represents the structured sequence of project activities required to develop the **Medicine Reminder & Tracker App**, along with their respective predecessor dependencies and durations:

No.	Activity	Predecessor(s)	Duration
1	Project Initiation & Requirement Analysis	—	1.0 Month
2	System Architecture & Technology Selection	1	0.5 Month
3	UI/UX Design (Elderly-Friendly & Bangla Layouts)	1	1.0 Month
4	Database Design & API Specification (SMS & Stock Tracking)	2, 3	1.0 Month
5	Backend Development (AI Reminders & Emergency Core Logic)	4	1.5 Months
6	Frontend Development (Native Android App Interface)	3, 4	1.5 Months
7	Core Feature Integration (Scanner, SMS Gateway & Voice Tools)	5, 6	1.0 Month

No.	Activity	Predecessor(s)	Duration
8	System Integration & End-to-End Testing	5, 6, 7	0.5 Month
9	Performance Optimization & Security Testing (Data Integrity)	8	0.5 Month
10	Deployment, User Training & Final Documentation	9	0.5 Month

Total Project Duration: 6 Months



6) Functional Requirements

User & Core Features

- **User Registration/Login:** Securely create and access personal health accounts to store prescription history.
- **Add Medicine:** Easily input medicine names, precise dosages, and specific time schedules.
- **Smart Reminder:** AI-based adaptive reminder notifications to ensure adherence.
- **Missed Dose Detection:** Automatically logs and records instances where a user misses their medication window.
- **Emergency Alert:** Immediately dispatches an automated SMS/message to registered family members if medication is missed more than once consecutively.
- **Prescription Scanner:** Uses optical camera scanning to extract details and auto-fill medicine schedules.
- **Medicine Stock Tracking:** Tracks real-time counts of remaining medicines and triggers warning notifications before a shortage occurs.
- **Emergency Doctor Consultation:** Provides a direct interface to instantly contact a medical professional during emergency health situations.

7) Non-Functional Requirements

- **Usability:** Highly simplified navigation layouts, optimized for elderly ergonomics with comprehensive Bangla language support.
- **Performance:** Ultra-fast operational response time and reliable, latency-free reminder triggers.
- **Reliability:** Core background reminder services must remain operational 24/7 without being closed by the OS background limits.
- **Security:** Safeguards user health data profiles via robust, secure authentication protocols.
- **Contact Protocols:** Features a dedicated, easily reachable directory for immediate emergency services.
- **Maintainability:** Formulated with modular code architecture to ensure smooth future updates and feature expansions.
- **Availability:** Data sync structures ensure that services and cached schedules remain accessible at any given time.
- **Data Integrity:** Implements validation logic to guarantee accurate, consistent, and corruption-free storage of vital medical metrics.

8) SDLC Model Selection

Selected Model: Agile Software Development Life Cycle (SDLC)

Justification

- **Continuous Feedback:** Allows our lecturer and project stakeholders to continuously evaluate progress over iterative sprints.
- **Risk Reduction:** Healthcare utility tools require rigorous testing; Agile allows for incremental testing of critical features (like SMS alerts and AI triggers) early on.
- **Adaptive Design:** Makes it easy to adapt the interface elements based on real usability testing with elderly individuals during the development phase.

9) Overall Conclusion

The Medicine Reminder & Tracker App UI/UX design successfully delivers a modern, accessible, and highly practical digital health solution tailored specifically to the Android platform and the Bangladesh family caregiving context. The Figma prototype comprises 15 fully designed screens covering the complete user journey for patients, elderly users, family caregivers, and emergency scenarios.

The design achieves the following key objectives:

1. **Complete Onboarding & Authentication Flow** – Separate, clear login/registration paths for patients and family members, with a dedicated accessibility setup for voice assistance.
2. **Full Patient Experience** – Patients can manually add medicines, scan prescriptions (OCR/QR), receive AI-driven smart reminders, mark doses as taken or skipped, track stock levels with color-coded warnings, and view missed dose history.
3. **Life-Saving Emergency Features** – One-tap SOS alert sends SMS to family members, missed dose escalation logic, and a direct emergency doctor contact interface.
4. **Family Caregiver Integration** – Family members can monitor adherence remotely, receive automated alerts, send reminders to the patient, and view 7-day compliance charts – perfectly aligned with the Bangladesh family structure.

5. **Accessibility First** – Voice reminders (Bangla/English), large touch targets, high-contrast options, and screen reader support make the app usable for elderly and visually impaired users.
6. **Prototype Interactions** – All 50+ buttons are connected with Smart Animate transitions, creating a fully interactive prototype that accurately simulates the final Android application.

The design is technically feasible (Android Studio, SMS API, OCR/QR libraries), operationally practical (simple UI, Bangla support), and economically sound with a Benefit–Cost Ratio of 3.0. The Agile design approach ensures flexibility for future enhancements such as pharmacy integration, wearables sync, and multi-language expansion beyond Bangla.

The **Medicine Reminder & Tracker App** represents a highly viable investment for improving medication adherence, reducing health risks for elderly patients, and providing peace of mind to families in Bangladesh.